Reviewer Comments

Recommendation: Major revision.

The core question is valid and unaddressed: when a multi-hop QKD network exhausts key, should it refuse service or serve at reduced assurance? The forced assurance loss metric is a reasonable addition. Concerns below are about evidence scope and framing, not simulator correctness.

Main concerns

1. **Single topology.** All headline numbers come from one nine-node graph. The authors state the attack budget equals the min-cut by definition, so 0.304 and 67.4% are properties of this graph, not general results. Evaluate on multiple topologies with varying cut size and report how the metric scales.
2. **Non-equivalent metrics.** Under fail-degraded, a session counts as available even when served with post-quantum crypto only. The 0.973 versus 0.304 comparison measures two different things. Promote the composite objective from §V-E to the primary comparison, or make the asymmetry explicit in the abstract.
3. **Validation overstated.** V1 compares the model against its own accounting; V4 bounds only to an order of magnitude. Two of four checks failed. State what each check establishes rather than claiming two reproductions.
4. **Wrong default parameter.** Field data implies n_auth of 7 to 40; the paper uses 2. Report main tables at the calibrated value with 2 as the optimistic sensitivity. The availability result is cut-limited, but the G1 affordability and §V-B routing results are not.
5. **Uniform attack cost.** The topology mixes fibre and free-space links, yet budget k assumes all links are equally attackable. Add per-link attack cost or justify the uniform assumption. Without this, the forcing cost is not a cost.
6. **Correlated outage affects the baseline.** For a mountain formation, weather or terrain masking could sever the min-cut with no adversary present. This weakens the claim that all loss at k > 0 is attack-attributable. Address the implication for the zero baseline.
7. **Metric ignores exposure.** A downgraded 2 kbit command message is weighted the same as a 320 kbit logistics transfer, although the fail-closed argument depends on differential harm. Add weighting by data sensitivity, or justify unweighted counting.
8. **Rescale the G1 class.** The one-time-pad floor is unfundable by three orders of magnitude, yet ISR dominates the key economy at 2880 of 2897 bit/s. A class that cannot be served under any configuration distorts the grade mix and the routing comparison. Rescale ISR to what the links can carry.
9. **Policy search is not required.** The useful output is the λ threshold, which follows from the objective definition. The cross-entropy search adds a component that does not strengthen the central claim, and "104.8%" bounds only the search budget. Compress §V-E and use the space for the topologies in point 1.

Minor points

- $\gamma = 3.4$ is fitted to one datapoint from a different QKD modality. Call it an illustrative bound, not a calibration.
- §V-B(1): the zero-width CI follows from the setup, not from measurement. Soften the wording.
- Table III needs confidence intervals; Table II needs a per-class breakdown and how SD ≤ 0.012 was computed.
- Table I: the all-✓ row for "This work" is self-serving. State the gap in prose.
- Fig. 1(b) is overloaded. Separate the two attack surfaces.
- Nearly every clause carries a caveat. Compress and use the space for the missing topology sweep.
- References [5] is dated 2026. Confirm publication status.

Questions

1. If ISR demand is scaled to what the free-space hops can sustain, does the fail-mode gap persist, narrow, or invert?
2. Does the capability/budget non-substitution hold for graphs with min-cut greater than 2?
3. Under an adversary that re-targets as the router reroutes, does the centrality-guided forcing cost still grow linearly with depth?
4. Would the classical-channel dependency change if the fallback used pre-placed symmetric keys instead of an online KEM?